

Notes: Guriev, Melnikov, and Zhuravskaya (QJE, 2021)

3G Internet and Confidence in Government

Contents

1	Big picture	3
2	Data and measurement (key objects)	4
2.1	Why 3G is the relevant technology	4
2.2	Gallup World Poll data and the global sample	4
2.3	Government approval and related outcomes	5
2.4	Measuring 3G coverage	5
2.5	Controls and development proxies	5
2.6	Mechanism data: censorship and corruption	6
2.7	Lightning strikes and the IV strategy	7
2.8	European election data	7
2.9	Descriptive facts and magnitudes	7
3	Models and empirical specifications (all equations + interpretation)	8
3.1	Mechanism and hypotheses	8
3.2	Baseline government-approval regression	8
3.3	Internet-at-home equation	9
3.4	Lightning IV first stage	9
3.5	Event-study design around sharp 3G expansions	9
3.6	2G placebo specification	10
3.7	European election regressions	10
3.8	Estimation choice and standard errors	10
3.9	What each equation is trying to learn	11
4	Identification and empirical design (what makes the estimates credible)	11
4.1	What identifies the baseline effect	11
4.2	Country-year fixed effects and local common shocks	11
4.3	Pretrends in leads and lags	11
4.4	Event-study around sharp expansions	12
4.5	2G as a placebo treatment	12
4.6	Omitted variables and selection on observables	12

4.7	Lightning IV	13
4.8	Placebo outcomes	13
4.9	What the paper can and cannot distinguish	13
5	Tables and figures	14
5.1	Figure I: the global 3G rollout	14
5.2	Table I and Figure II: the baseline relationship	15
5.3	Table II and Figures III–IV: pretrends and the event study	16
5.4	Table III and Table IV: placebo 2G and lightning IV	16
5.5	Table V and Figure V: censorship of the internet and traditional media	18
5.6	Figure VI, Table VI, Table VII, and Figure VII: the corruption mechanism	18
5.7	Table VIII, Table IX, and Figure VIII: electoral implications in Europe	20
6	Short summary	22
7	Conclusion	22
7.1	Core conclusion (what the paper establishes)	22
7.2	Economic intuition	22
7.3	Takeaway	22

1 Big picture

- **Question.** How does the expansion of mobile broadband internet affect confidence in government, perceptions of corruption, and ultimately electoral outcomes?
- **Setting.** The paper studies the global rollout of 3G mobile networks using Gallup World Poll micro data on roughly **840,537 individuals** in **2,232 subnational regions** across **116 countries** from **2008 to 2017**. For elections, it studies **102 parliamentary elections** in **33 European democracies** at the district level.
- **Core challenge.** 3G rollout was not random. Regions receiving coverage earlier may also differ in development, political trends, media environment, or local economic shocks. A simple cross section would confound mobile internet with many omitted factors.
- **Main conceptual contribution.** The paper is the first global study of the political effects of **mobile broadband internet** rather than older mobile technology. Its main innovation is to connect 3G access to government approval, censorship, actual corruption exposure, and electoral support for incumbents and populists.
- **Main hypotheses.**
 - Uncensored mobile broadband gives citizens access to independent political information and lowers government approval when it reveals poor governance.
 - The effect should be stronger where traditional media are censored, because the internet becomes a more important news source.
 - The effect should be stronger when there is actual corruption to expose.
 - Earlier mobile technology such as 2G should not have the same effect, because it does not allow easy browsing, video sharing, and social-media use.
- **Main empirical design.** The baseline design is a difference-in-differences panel that compares changes in government approval across subnational regions as 3G coverage expands, controlling for region fixed effects, year fixed effects, and a large set of covariates. Credibility is reinforced with country-year fixed effects, event studies around sharp 3G expansions, a 2G placebo, and an IV that uses lightning strikes to predict the speed of rollout.
- **Main baseline result.** More 3G coverage lowers government approval. A decade-long average increase in regional 3G coverage of **0.39** lowers confidence in the national government by about **2.5 percentage points**, increases perceived corruption by about **1.4 percentage points**, and lowers the aggregate government-approval principal component by about **2.2 percentage points**.
- **Key heterogeneity result.** The negative effect appears only where the internet is **not censored**, and it is stronger where traditional media are **more censored**. It is also stronger for **rural residents**, the unemployed, lower-income and less-educated respondents, and weaker for the young.
- **Key mechanism result.** 3G strengthens the link between **actual corruption** and **perceived corruption**. It therefore seems to help citizens learn about real corruption, not just change their mood.

- **Key electoral result.** In Europe, an average decade-long district increase in 3G coverage of **0.53** lowers incumbent vote share by about **4.7 percentage points**, lowers turnout by about **2 percentage points**, and raises the vote share of populist parties, especially right-wing and left-wing populists.
- **Main credibility result.** The results survive country-year fixed effects, show no pretrends, survive an event-study design around sharp 3G expansions, pass a 2G placebo test, remain strong in an IV specification using lightning strikes, and do not show up for placebo outcomes like life satisfaction or confidence in the local police.
- **Economic takeaway.** Mobile broadband internet can undermine governments by exposing misgovernance and corruption, especially when it provides the first important source of uncensored political information. In democracies, the same shock can also shift electoral support away from incumbents and toward antiestablishment populists.

2 Data and measurement (key objects)

2.1 Why 3G is the relevant technology

- The paper focuses on **3G mobile internet**, the first generation of mobile technology that allowed users to browse the web freely from smartphones and stream or upload videos.
- This matters because 3G was a major technological precondition for large-scale social-media use.
- The paper treats **2G** as an explicit placebo technology: 2G allowed calls and texts, but only very limited internet use.

Interpretation. The paper is not about telecommunications infrastructure in general. It is about the arrival of a specific communication technology that sharply changes the production and diffusion of political information.

2.2 Gallup World Poll data and the global sample

- The core survey data come from the **Gallup World Poll**.
- The main sample covers **840,537 respondents** in **13,004 region-year cells**, drawn from **2,232 subnational regions** in **116 countries**.
- The mean number of times a region appears is about **6**; more than **75%** of regions appear in at least four years.
- On average, there are about **65 respondents** per region-year.
- Approximately **80%** of interviews are face-to-face and **20%** are by phone.

Interpretation. The paper has both enormous geographic coverage and repeated within-region variation over time, which is exactly what the identification strategy needs.

2.3 Government approval and related outcomes

The Gallup survey contains four political outcomes:

- confidence in the **national government**,
- confidence in the **judicial system and courts**,
- confidence in the **honesty of elections**,
- whether **corruption is widespread in government**.

The paper uses these four outcomes in three ways:

- separately, one by one;
- as the **share of positive answers**;
- as the **first principal component** of the four responses.

The survey also includes a dummy for **internet access at home**. **Interpretation.** The authors deliberately examine several dimensions of political trust instead of relying on a single broad approval question.

2.4 Measuring 3G coverage

- The paper uses digital maps of global **3G network coverage** from **2007 to 2018**, based on mobile-operator submissions aggregated by the GSM Association.
- Coverage is observed on a **1 km by 1 km grid**.
- If a grid cell is covered by 4G, it is also covered by 3G by definition.
- The 2011 data are missing because of a change in data administration, so the paper interpolates 2011 coverage using the mean of 2010 and 2012.

For region r in year t , regional 3G coverage is a population-weighted average of grid-cell coverage:

$$3G_{rt} = \sum_{g \in r} \omega_{gr} \cdot 1\{3G \text{ available in grid cell } g \text{ in year } t\}, \quad (1)$$

where the weights ω_{gr} come from local population density and sum to 1 within the region.

Interpretation. The paper wants the share of people, not the share of raw land, covered by mobile broadband.

2.5 Controls and development proxies

The baseline regressions include:

- age and age squared,
- gender,

- marital status,
- education dummies,
- employment status,
- urban or rural residence,
- the region’s average level of household income,
- the country’s log GDP per capita,
- the country’s unemployment rate,
- and democracy indicators.

The main measure of regional development is the log of **mean household income** among respondents in the region. The paper shows robustness to using **nighttime lights** instead. **Interpretation.** These controls matter because 3G may arrive first in richer, faster-growing, or more democratic places.

2.6 Mechanism data: censorship and corruption

To study mechanisms, the paper adds several country-level variables.

Internet censorship:

- Freedom House *Freedom on the Net* **Limits on Content** score;
- both time-varying and time-invariant versions;
- high-censorship dummies are built from thresholds in the score distribution.

Traditional-media censorship:

- Freedom House *Freedom of the Press* score;
- again used in both time-varying and time-invariant forms.

Actual corruption:

- the IMF’s **Global Incidents of Corruption Index** (GICI), based on text analysis of Economist Intelligence Unit country reports;
- a country-level overall-corruption measure equal to the long-run mean of the GICI from 2000 to 2017;
- the number of country-specific entities named in the **Panama Papers**, both in levels and per capita.

Interpretation. These variables let the authors move beyond “3G lowers approval” and ask *why* it does so.

2.7 Lightning strikes and the IV strategy

The instrument comes from the **World Wide Lightning Location Network**.

- The paper calculates the average annual number of cloud-to-ground lightning strikes in each region between **2005 and 2011**.
- Lightning is weighted by population density and divided by regional area.
- A region is “high lightning” if it lies in the top half of the global distribution.

Interpretation. Frequent lightning increases infrastructure costs by creating power surges, slowing network rollout, especially in poorer countries.

2.8 European election data

For the electoral analysis, the paper assembles:

- **102 parliamentary elections**,
- in **33 European countries**,
- across **398 subnational districts**,
- yielding **1,250 district-election observations**.

The main electoral outcomes are:

- vote share of the **top two parties from the first election**,
- vote share of the **ruling party**,
- turnout,
- vote share of **right-wing populists**,
- vote share of **left-wing populists**,
- vote share of **all populists**,
- vote share of **Green parties**,
- and vote share of **nonpopulist opposition parties**.

Interpretation. The election data allow the paper to connect survey attitudes to actual political behavior.

2.9 Descriptive facts and magnitudes

- Mean regional 3G coverage in the main global sample is about **0.38**.
- The average increase in 3G coverage across regions from 2008 to 2017 is about **0.39**.

- In Europe, the average increase in district 3G coverage over the 2008–2017 decade is about **0.53**, roughly from **37%** to **90%**.
- In the baseline sample, the effect of an average decade-long 3G expansion is comparable in size to about a **2.2 percentage point** rise in the national unemployment rate.

Interpretation. The estimated effects are not tiny. They are politically meaningful relative to familiar macro shocks.

3 Models and empirical specifications (all equations + interpretation)

3.1 Mechanism and hypotheses

The paper’s mechanism is fundamentally informational:

- 3G raises access to uncensored political information, especially social media and online video;
- that information can expose actual corruption and poor government performance;
- the effect should be stronger where no strong alternative source of independent information exists;
- the effect may also empower antiestablishment politicians.

The paper is careful not to claim that the internet only spreads true information. It explicitly allows that **false narratives** may also matter, especially for populist gains. **Interpretation.** The model behind the regressions is a reduced-form information shock, not a fully structural persuasion model.

3.2 Baseline government-approval regression

The main individual-level specification is:

$$\text{GovApproval}_{irt} = \gamma_1 3G_{rt} + \gamma_2 \text{Development}_{rt} + X'_{irt} \lambda + \phi_r + \tau_t + \varepsilon_{irt}, \quad (2)$$

where:

- i indexes individuals,
- r indexes subnational regions,
- t indexes years,
- $3G_{rt}$ is the share of the regional population covered by 3G,
- ϕ_r are region fixed effects,
- τ_t are year fixed effects,

- and X_{irt} is the vector of individual and macro controls.

Interpretation.

- Region fixed effects absorb all time-invariant differences across places.
- Year fixed effects absorb worldwide time shocks.
- The coefficient γ_1 is identified from changes in 3G coverage within regions over time.

3.3 Internet-at-home equation

Because the Gallup data do not observe actual social-media use, the paper estimates a reduced-form “internet access” equation:

$$\text{InternetAtHome}_{irt} = \alpha_1 3G_{rt} + \alpha_2 \text{Development}_{rt} + X'_{irt} \lambda + \phi_r + \tau_t + \varepsilon_{irt}. \quad (3)$$

Interpretation. This is only a partial test of the extensive margin:

- it does not observe internet use outside the home,
- it does not distinguish broadband from slow access,
- and it does not capture the qualitative change in how people use the internet after 3G.

3.4 Lightning IV first stage

To address endogeneity of rollout, the paper estimates a first-stage equation at the region-year level:

$$3G_{rt} = \delta_1 [\text{Lightning}_r \times t \times \text{Rich}_{cr}] + \delta_2 [\text{Lightning}_r \times t \times \text{Poor}_{cr}] + Z'_{rt} \mu + \phi_r + \tau_t + \varepsilon_{rt}, \quad (4)$$

where:

- Lightning_r indicates high lightning-strike frequency in region r ,
- Rich_{cr} and Poor_{cr} indicate whether the region’s country is above or below median GDP per capita,
- Z_{rt} includes baseline controls and additional interacted linear time trends.

The second stage replaces actual $3G_{rt}$ with predicted 3G coverage. **Interpretation.** The first stage is expected to be strongest in poorer countries where surge protection is scarcer and more expensive.

3.5 Event-study design around sharp 3G expansions

The paper defines an “event” as an annual increase in regional 3G coverage of more than **50 percentage points**. It then estimates dynamic effects using year indicators relative to the event.

Interpretation.

- This is a very sharp treatment.
- It lets the authors ask whether approval falls *before* 3G arrives or only *after*.
- It is especially useful because ordinary leads of 3G can be mechanically correlated with current 3G due to persistent rollout.

3.6 2G placebo specification

The paper also estimates analogues of the baseline equation replacing 3G coverage with **2G coverage**, and in some regressions includes both:

$$\text{GovApproval}_{irt} = \beta_1 3G_{rt} + \beta_2 2G_{rt} + X'_{irt}\theta + \phi_r + \tau_t + \varepsilon_{irt}. \quad (5)$$

Interpretation. If the estimated effect were just “new telecom infrastructure,” 2G should look similar to 3G. If the mechanism is internet-based information diffusion, 2G should not look similar.

3.7 European election regressions

For elections, the paper estimates district-election versions of the baseline idea:

$$\text{VoteShare}_{dt} = \beta 3G_{dt} + W'_{dt}\theta + \delta_d + \tau_t + u_{dt}, \quad (6)$$

with:

- district fixed effects,
- year fixed effects,
- district income proxied by nighttime lights,
- and country-level macro controls such as GDP per capita, unemployment, inflation, labor-force participation, and population age structure.

For ruling-party regressions, the paper sometimes uses **incumbent-party-by-district fixed effects**. **Interpretation.** The election regressions ask whether regions that got more 3G support changed vote shares, not just whether countries with high internet penetration behave differently.

3.8 Estimation choice and standard errors

- The baseline government-approval regressions are estimated at the individual level by OLS.
- Standard errors are corrected for **two-way clustering** at the subnational-region level and the country-year level.
- The paper also shows robustness to country clustering and to spatially correlated standard errors.

- The IV specifications are estimated at the **region-year** level because both the instrument and treatment vary at that level.

Interpretation. The design is not driven by a fragile standard-error convention.

3.9 What each equation is trying to learn

- Baseline approval equation: Does more 3G lower approval?
- Internet-at-home equation: Does 3G really raise internet access?
- Lightning IV: Is rollout speed plausibly exogenous to politics?
- Event study: Are there pretrends?
- 2G placebo: Is this a broadband-specific political effect?
- Election regressions: Do approval changes translate into vote changes?

4 Identification and empirical design (what makes the estimates credible)

4.1 What identifies the baseline effect

The baseline estimate comes from **within-region changes** in 3G coverage over time:

- region fixed effects remove permanent regional differences;
- year fixed effects remove worldwide shocks;
- the remaining variation is the timing and intensity of rollout across regions.

Interpretation. The paper does not compare Finland to Brazil or cities to villages in levels. It compares the same region to itself before and after additional 3G arrives.

4.2 Country-year fixed effects and local common shocks

A strong concern is that approval and 3G could trend together for country-level reasons. The paper therefore adds **country-year fixed effects**, so identification comes only from differential 3G expansion across regions within the same country-year.

Interpretation. This is a demanding test because 3G often expands nationally, leaving much less variation. The fact that the main results largely survive strengthens the causal reading.

4.3 Pretrends in leads and lags

The paper estimates leads and lags of 3G coverage and finds:

- future 3G expansion does **not** predict current approval;
- past 3G expansion does predict lower approval;
- with country-year fixed effects, the coefficients on the leads are significantly smaller in absolute value than the contemporaneous effect.

Interpretation. Regions do not look politically different *before* getting 3G in a way that mirrors the estimated post-arrival effect.

4.4 Event-study around sharp expansions

The event study defines treatment as a one-year rise in regional 3G coverage of more than **50 percentage points**. There are **452 regions** in **65 countries** with such events.

The findings are:

- no meaningful pre-event movement in approval;
- a clear drop in approval immediately after the event;
- similar or stronger effects for rural residents.

The post-event dummy is about **-0.036** in the full sample and **-0.052** for rural residents. **Interpretation.** The event-study evidence is a clean visual counterpart to the baseline DID results.

4.5 2G as a placebo treatment

The 2G placebo is powerful because it uses nearly the same type of infrastructure but not the same information technology.

- 2G does **not** increase internet access at home.
- 2G does **not** lower government approval; if anything, it is positively related to some approval measures.
- Controlling for 2G does not materially change the 3G coefficient.

Interpretation. The effect is not “people like or dislike new towers.” It is specific to the informational content enabled by mobile broadband.

4.6 Omitted variables and selection on observables

The paper uses two standard approaches:

- **Altonji–Elder–Taber**-style logic using predicted 3G from observables;
- **Oster** bounds on how strong unobservables would need to be to overturn the results.

The estimated Oster δ values are generally large in magnitude or of the wrong sign to make omitted variables a plausible complete explanation. **Interpretation.** Observable selection patterns do not look like what would be needed for the entire 3G effect to be spurious.

4.7 Lightning IV

The IV strategy uses exogenous weather-related infrastructure costs:

- high lightning frequency slows 3G rollout, especially in poorer countries;
- the first-stage F -statistic is around **21** in the full sample and around **25** in poorer-country rural subsamples;
- the second-stage estimates remain negative and are larger in magnitude than OLS.

For the aggregate approval principal component:

- IV coefficient is about **-0.283** in the full sample,
- about **-0.329** in countries below median GDP per capita.

Interpretation. The IV confirms the sign and strengthens the argument that rollout timing matters causally.

4.8 Placebo outcomes

The paper also checks outcomes that should not move if the mechanism is specifically political information:

- life satisfaction today,
- expected life satisfaction in five years,
- satisfaction with current standards of living,
- beliefs that living standards are improving,
- confidence in the local police.

3G has little or no effect on these placebo outcomes. **Interpretation.** The main result is not just generalized pessimism.

4.9 What the paper can and cannot distinguish

The paper can establish that:

- mobile broadband lowers government approval on average;
- the effect depends on censorship and corruption;
- the effect translates into electoral losses for incumbents and gains for populists in Europe.

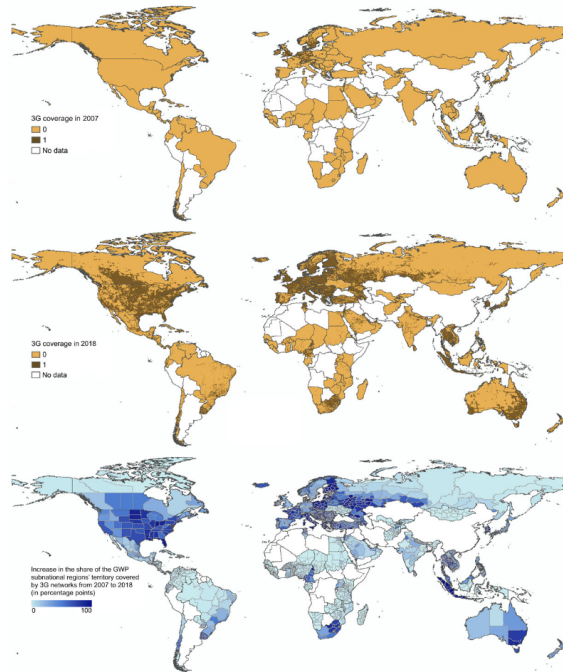


FIGURE I

The Growth of 3G Network Coverage between 2007 and 2018

The first two maps present 3G network coverage by grid cell in 2007 and 2018. The third map presents: (i) the boundaries of the subnational regions (the unit of localization in the GWP data) and (ii) the increase in the share of each subnational region's population covered by 3G networks from 2007 to 2018. The sample consists of all countries covered by the GWP data. There are 2,232 subnational regions in the sample.

But it cannot fully separate:

- truthful exposure of corruption from the spread of false narratives,
- all channels behind populist gains,
- or pure information from all other psychological and social-media mechanisms.

Interpretation. The paper is strongest as a disciplined reduced-form political-economy study, not as a complete theory of social media and democracy.

5 Tables and figures

5.1 Figure I: the global 3G rollout

Read:

- Figure I maps 3G coverage in **2007** and **2018**, then maps the increase in regional population coverage over the whole period.
- The map makes clear that the paper is exploiting large cross-region differences in timing and intensity of rollout.

- The sample contains **2,232** subnational regions worldwide.

Economic message. The identifying treatment is genuinely global, not driven by one country or one continent.

5.2 Table I and Figure II: the baseline relationship

TABLE I
THE EFFECT OF MOBILE INTERNET COVERAGE ON CONFIDENCE IN GOVERNMENT

Dep. var.:	Confidence in national government (1)	Confidence in judicial system (2)	Honesty of elections (3)	No corruption in government (4)	Share of questions with positive responses (5)	1st principal component of responses (6)
Panel A: Sample of all respondents						
Regional 3G coverage	-0.063*** (0.021)	-0.040*** (0.015)	-0.079*** (0.021)	-0.036** (0.014)	-0.056*** (0.015)	-0.057*** (0.015)
Observations	772,353	748,471	732,856	722,768	617,863	617,863
R-squared	0.164	0.163	0.168	0.225	0.242	0.239
Mean dep. var	0.439	0.534	0.505	0.226	0.432	0.439
Mean 3G coverage	0.397	0.381	0.383	0.383	0.381	0.381
Number of countries	111	116	112	112	110	110
Panel B: Subsample of rural residents						
Regional 3G coverage	-0.091*** (0.024)	-0.058*** (0.017)	-0.115*** (0.026)	-0.054*** (0.016)	-0.089*** (0.018)	-0.081*** (0.018)
Observations	464,831	448,449	440,786	432,460	371,055	371,055
R-squared	0.171	0.157	0.161	0.194	0.224	0.222
Mean dep. var	0.349	0.556	0.516	0.215	0.445	0.452
Mean 3G coverage	0.329	0.314	0.316	0.316	0.311	0.311
Number of countries	110	115	111	111	109	109
Subnational region and year FEs	✓	✓	✓	✓	✓	✓
Baseline controls	✓	✓	✓	✓	✓	✓

Notes: *** $p < .01$, ** $p < .05$, * $p < .1$. 3G internet reduces government approval. The unit of observation is an individual. Panel A reports the results for the full sample and Panel B for the subsample of respondents from rural areas. The table presents the results of the estimation of specification (1). The dependent variables are individuals' perceptions of government and the country's institutions. Controls include age, sex, regional, gender, marital status, duration for high school and university education, employment status, urban status, the region's average level of income, the log of the country's GDP per capita, the country's unemployment rate, and duration for democracy status. Standard errors in parentheses are corrected for two-way clusters at the level of the subnational regions (to account for correlation over time) and at the level of the countries in each year (to account for within-country-year correlation).

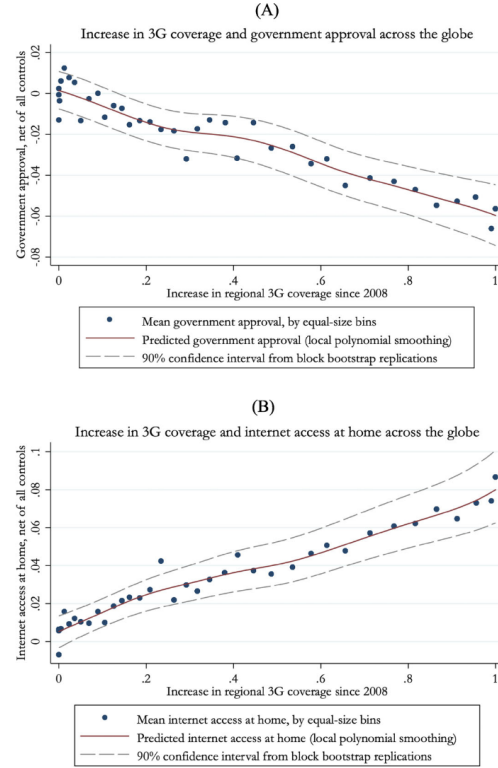


FIGURE II

Increase in 3G Coverage and Confidence in Government

Panel A illustrates the relationship between regional 3G coverage and government approval (Table I, Panel A, column (6)). Panel B of the figure illustrates the relationship between the increase in regional 3G coverage and access to the internet at home (Online Appendix Table A.2, Panel A, column (1)). The dots show the means of the respective outcome variables, net of all controls, by equal-size bins. The lines on the graphs show the predicted outcomes (Gaussian kernel, local polynomial smoothing). The confidence intervals are constructed by performing a block bootstrap at the level of the clusters.

Read:

- Table I is the core baseline result.
- In the full sample, the coefficient on regional 3G coverage is about **-0.063** for confidence in the national government, **-0.040** for confidence in the judiciary, **-0.079** for confidence in election honesty, **-0.036** for the “no corruption” outcome, and **-0.057** for the first principal component of approval.
- In the rural sample, the coefficients are larger in absolute value: for example, **-0.091** for national government confidence and **-0.081** for the principal component.
- Figure II shows a smooth negative relationship between the increase in 3G coverage and resid-

ualized government approval, and a positive relationship between 3G expansion and internet access at home.

Economic message. Mobile broadband expands internet access but lowers support for government. The effect is stronger in rural areas.

5.3 Table II and Figures III–IV: pretrends and the event study

Dep. var.:	1st principal component of the government approval responses					
Sample note 1:	Regions with a sharp increase in 3G coverage in one year in 2007–2018					
Sample note 2:	All respondents			Rural respondents		
	(1)	(2)	(3)	(4)	(5)	(6)
Regional 3G coverage	-0.055*** (0.014)			-0.073*** (0.018)		
Postevent dummy		-0.036*** (0.012)			-0.052*** (0.014)	
Sharp increase in regional 3G coverage occurred in:						
Year $t + 4$ or later			-0.015 (0.021)			-0.011 (0.024)
Year $t + 3$			0.001 (0.019)			-0.005 (0.021)
Year $t + 2$			-0.006 (0.014)			0.006 (0.017)
Year t			-0.032** (0.013)			-0.035** (0.015)
Year $t - 1$			-0.048*** (0.016)			-0.066*** (0.019)
Year $t - 2$			-0.033* (0.019)			-0.053** (0.021)
Year $t - 3$			-0.063*** (0.022)			-0.067*** (0.024)
Year $t - 4$ or earlier			-0.051* (0.030)			-0.061** (0.030)

Dep. var.:	1st principal component of the government approval responses					
Sample note 1:	Regions with a sharp increase in 3G coverage in one year in 2007–2018					
Sample note 2:	All respondents			Rural respondents		
	(1)	(2)	(3)	(4)	(5)	(6)
Observations	130,406	130,406	130,406	66,078	66,078	66,078
R-squared	0.213	0.212	0.213	0.242	0.242	0.242
Number of countries	65	65	65	62	62	62
Number of regions	452	452	452	444	444	444
Number of countries with variation	65	36	65	62	32	62
Number of regions with variation	452	219	452	444	206	444
Subnational region and year FEs	✓	✓	✓	✓	✓	✓
Baseline controls	✓	✓	✓	✓	✓	✓
Censorship of the traditional press control	✓	✓	✓	✓	✓	✓
p-value: $\chi^2(Y_t) = \chi^2(Y_{t-1})$			.119			.010
p-value: $\chi^2(Y_{t+1}) = \chi^2(Y_{t-1})$			.024			.005

Notes. *** $p < .01$, ** $p < .05$, * $p < .1$. The table presents the results of the event study. The unit of observation is an individual. The sample is comprised of individuals from regions that had a sharp increase in 3G coverage, more than 10 percentage points in the share of a subnational region's population covered by 3G in a single year, between 2007 and 2018. There are 452 regions from 65 countries like this. All regions in this sample have variation in the lag and leads of the year of the event (estimated in columns (1) and (6)). However, only 219 regions out of all regions with an event have variation in the postevent dummy within the sample, due to missing region-years in GWP data. Columns (1)–(3) report results for the full sample; columns (4)–(6) show results for the subsample of respondents from rural areas. The unreported controls include age, age squared, gender, marital status, dummies for high school and university education, employment status, urban status, the region's average level of income, the lag of the country's GDP per capita, the country's unemployment rate, dummies for democracy status, and ownership-of-the-traditional-press score. Standard errors in parentheses are corrected for two-way clusters at the level of the subnational regions (to account for correlation over time) and at the level of the countries in each year (to account for within-country-year correlation).

Read:

- Figure III shows that leads of 3G do not predict current approval once country-year fixed effects are included, whereas lags remain negative.
- Table II studies regions with a sharp increase in 3G coverage of more than **50 percentage points** in one year.
- The post-event dummy is about **-0.036** in the full sample and **-0.052** for rural residents.
- Figure IV shows no visible pretrend before the event and a clear fall in government approval after the event.

Economic message. The timing evidence looks like treatment arriving and then approval falling, not approval falling first and rollout following.

5.4 Table III and Table IV: placebo 2G and lightning IV

Read:

- Table III shows that **2G** does not increase internet access at home and does not generate the same negative government-approval effect as 3G.
- In the combined regression, 3G still has a coefficient of about **-0.053** on the approval principal component, while 2G is **+0.048**.
- Table IV reports the IV results. For the approval principal component, the IV coefficient is about **-0.283** in the full sample and **-0.329** in countries below median GDP per capita.
- The first-stage F -statistics are around **20.7**, **19.1**, **25.2**, and **25.5** depending on the sample.

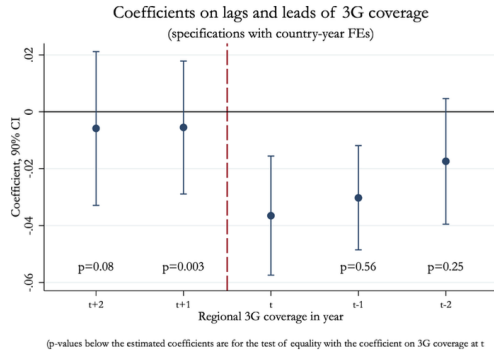


FIGURE III
Pretrend Analysis with Country $\times$ Year FEs

The figure presents the coefficients from the regressions of government approval on the lags and leads of 3G coverage in the full sample, controlling for country-year fixed effects and all the baseline controls. Each coefficient is from a separate regression. The results suggest that future expansions of 3G networks are not associated with current changes in government approval. The p -values below the estimates are for the test of equality of magnitudes between the respective coefficient and the coefficient on regional 3G coverage at t . The coefficients on the leads of 3G coverage are significantly smaller in absolute value than on 3G coverage at t , confirming the parallel pretrends assumption required for identification.

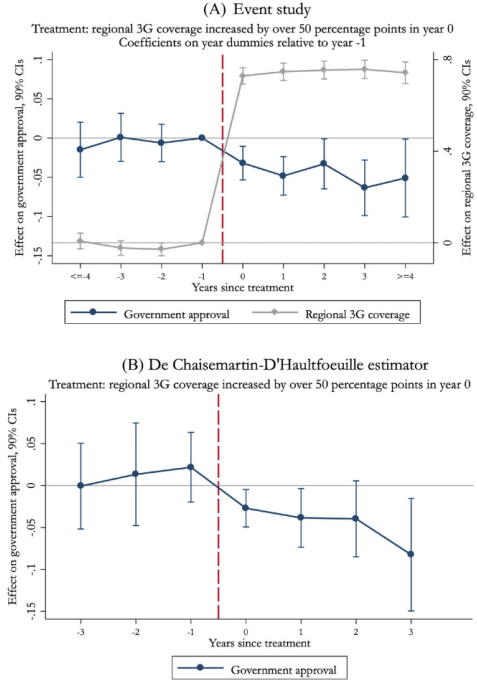


FIGURE IV
Event Study Analysis

Panel A presents an event study in which government approval (left axis) and 3G coverage (right axis) are regressed on a set of year dummies around the event defined as an annual increase in regional 3G coverage of more than 50 percentage points. The regressions are run on the subsample of 452 regions in 65 countries where 3G did increase sharply in a single year between 2007 and 2018. The results of the underlying regression for government approval as outcome are presented in Table II, column (3). For each outcome variable, all the coefficients come from the same regression, which includes all the baseline controls and the freedom-of-the-press score in the list of covariates. Panel B presents the estimates based on the estimator proposed in De Chaisemartin and D'Haultfoeuille (2020), which ensures that the average treatment effects in each group and period do not have negative weights. Both panels show that the decrease in government approval occurred after the significant expansion of 3G networks.

Dep. var.:	Confidence in national government	Confidence in judicial system	Honesty of elections	No corruption in government	Share of questions with positive responses	1st principal component of responses	Internet access at home
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: The effect of 2G on confidence in government and internet access at home							
Regional 2G coverage	0.045 (0.029)	0.031 (0.020)	0.098*** (0.030)	0.054*** (0.019)	0.056*** (0.021)	0.056** (0.022)	-0.013 (0.020)
Observations	772,353	748,471	732,856	722,768	617,863	617,863	840,537
Mean dep. var.	0.514	0.534	0.505	0.226	0.432	0.439	0.440
Panel B: The effect of 3G and 2G on confidence in government and internet access at home							
Regional 3G coverage	-0.060*** (0.020)	-0.038*** (0.015)	-0.074*** (0.020)	-0.032** (0.014)	-0.053*** (0.015)	-0.053*** (0.015)	0.080*** (0.017)
Regional 2G coverage	0.037 (0.028)	0.026 (0.019)	0.088*** (0.030)	0.049** (0.019)	0.048** (0.021)	0.048** (0.021)	-0.002 (0.019)
Observations	772,353	748,471	732,856	722,768	617,863	617,863	840,537
Mean dep. var.	0.514	0.534	0.505	0.226	0.432	0.439	0.440
Subnational region and year FEs	✓	✓	✓	✓	✓	✓	✓
Baseline controls	✓	✓	✓	✓	✓	✓	✓

Notes: *** $p < .01$, ** $p < .05$, * $p < .1$. The table presents the effects of 2G coverage on internet usage and government support. The results suggest that, as expected, the change in 2G coverage did not increase internet access at home and, on average, increased government support. The unit of observation is an individual. Panel A reports results for the effect of 2G coverage. Panel B shows similar results with 3G coverage included as a control variable. Columns (1)–(6) present the results for government approval as the outcome variable; column (7) presents the results for a dummy for having access to the internet at home. Baseline controls include age, age squared, gender, marital status, dummies for high school and university education, employment status, urban status, the region's average level of income, the lag of the country's GDP per capita, the country's unemployment rate, and dummies for democracy status. Standard errors in parentheses are corrected for two-way clusters at the level of the subnational regions (to account for correlation over time) and at the level of the countries in each year (to account for within-country-year correlation).

Dep. var.:	Regional 3G coverage	1st principal component of government approval	Regional 3G coverage	1st principal component of government approval	Regional 3G coverage	1st principal component of government approval	Regional 3G coverage	1st principal component of government approval
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Stage, 2SLS:								
Countries in the sample:	All	All	Rural	Rural	All	All	Rural	Rural
Respondents in the sample:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Regional 3G coverage	-0.282*** (0.080)	-0.308*** (0.080)	-0.308*** (0.080)	-0.308*** (0.080)	-0.329*** (0.080)	-0.329*** (0.080)	-0.329*** (0.080)	-0.329*** (0.080)
1/High frequency of lightning strikes per sq. km. $\times$ Year $\times$ 1/2GDP per capita below median	-0.032*** (0.005)	-0.032*** (0.005)	-0.032*** (0.005)	-0.032*** (0.005)	-0.032*** (0.005)	-0.032*** (0.005)	-0.032*** (0.005)	-0.032*** (0.005)
1/High frequency of lightning strikes per sq. km. $\times$ Year $\times$ 1/2GDP per capita above median	-0.010*** (0.005)	-0.010*** (0.005)	-0.010*** (0.005)	-0.010*** (0.005)	-0.010*** (0.005)	-0.010*** (0.005)	-0.010*** (0.005)	-0.010*** (0.005)
Observations	12,860	12,860	11,743	11,743	5,789	5,789	5,324	5,324
Mean dep. var.	0.373	0.432	0.369	0.439	0.154	0.433	0.154	0.440
F-stat, excluded instrument	20.74	20.74	19.15	19.15	25.15	25.15	25.15	25.15
Corresponding OLS coefficient on regional 3G coverage					-0.129*** (0.027)	-0.129*** (0.027)	-0.129*** (0.027)	-0.129*** (0.027)
Subnational region and year FEs	✓	✓	✓	✓	✓	✓	✓	✓
Extended set of controls	✓	✓	✓	✓	✓	✓	✓	✓

Notes: *** $p < .01$, ** $p < .05$, * $p < .1$. The table presents the results of an IV analysis, where the frequency of lightning strikes per sq. km in a subnational region is used as an IV for the expansion of regional 3G coverage. The methodology follows Mansueto and Tsai (2020). High frequency of lightning strikes per sq. km is defined by the subnational region being in the top half of the distribution of lightning strikes per sq. km. Odd columns present the results of the first stage. Even columns show the results of the second stage. Columns (1)–(4) present the results for all the countries in the sample; columns (5)–(8) show results for the subsamples of countries with below median GDP per capita. The unit of observation is a subnational region. Controls include the region's average level of income, the lag of the country's GDP per capita, the country's unemployment rate, dummies for democracy status, and linear time trends interacted with the subnational regions' share of territory covered by forests, share of territory covered by mountains, maximum elevation, dummies for each quintile of population density, 3G coverage in 2008, a dummy for whether the region had any 3G coverage in 2008, and a dummy for whether the country had any 3G coverage in 2008. Standard errors in parentheses are corrected for two-way clusters at the level of the subnational regions (to account for correlation over time) and at the level of the countries in each year (to account for within-country-year correlation).

5.5 Table V and Figure V: censorship of the internet and traditional media

Notes: *** $p < .01$, ** $p < .05$, * $p < .1$. The unit of observation is an individual. The dependent variables are individuals' perceptions of government and the country's institutions. Panels A and C use time-varying measures of membership, whereas Panels B and D use time-invariant measures. Unreported controls include age, age squared, gender, marital status, distance to high school and university education, employment status, urban status, the region's average level of income, the log of the country's GDP per capita, the country's unemployment rate, and dummies for democracy status. Standard errors in parentheses are corrected for two-way clusters at the level of the educational region. The access

- In Table V, when the internet is not censored, 3G coverage has a strong negative effect on all approval measures.
- The interaction between 3G and internet censorship is positive and roughly offsets the direct negative 3G effect.
- When traditional media are more censored, the 3G effect becomes *more* negative.
- Figure V shows that 3G increases internet access at home in both censored and uncensored countries, but government approval falls only where the internet is uncensored.

Economic message. The results are not about connectivity alone; they are about access to *independent* political information.

5.6 Figure VI, Table VI, Table VII, and Figure VII: the corruption mechanism

- Figure VI groups countries by overall corruption and shows that the average 3G effect is negative almost everywhere, except in the least corrupt group.
- The least corrupt group consists of **Denmark, Germany, Japan, the Netherlands, Norway, Sweden, Switzerland, and the United Kingdom**; in that group, 3G expansion increases government approval.
- Table VI shows that 3G strengthens the link between **actual corruption incidents** (GICI) and perceived corruption. In places with no 3G, the relation is weak; in places with full 3G coverage, it is large and negative.
- A one-standard-deviation increase in the GICI is associated with about a **2.9 percentage point** fall in the perceived cleanliness of government in fully covered regions, versus only about **0.4 percentage points** in uncovered regions.

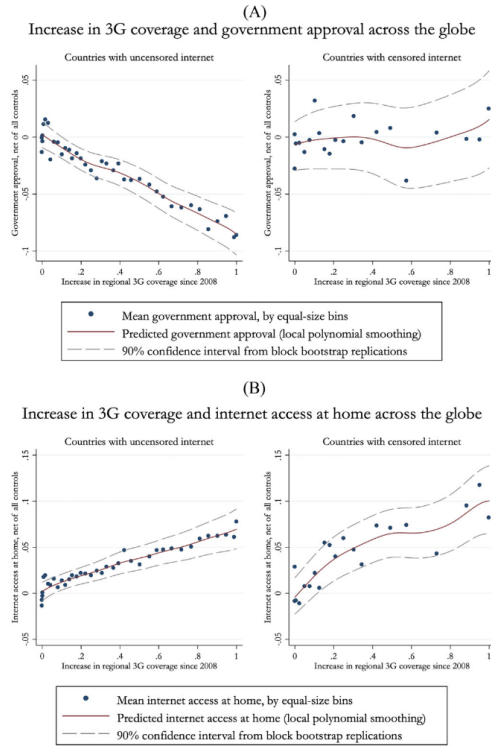
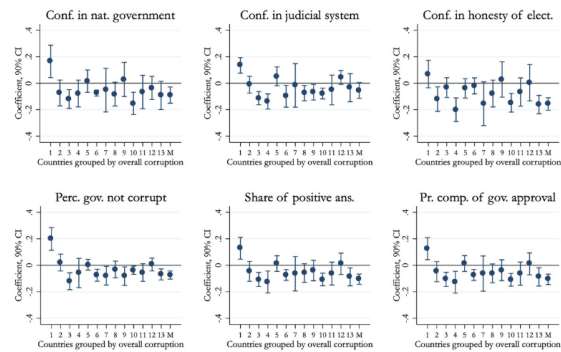


FIGURE V

Increase in 3G Coverage and Confidence in Government, Depending on Internet Censorship

Panel A illustrates the results presented in Table V, Panel B, column (6), showing the relationship between the increase in regional 3G coverage and government approval separately for countries with and without internet censorship. Panel B illustrates the relationship between the increase in regional 3G coverage and access to the internet at home for countries with high and low levels of internet censorship. The dots show the means of the respective outcome variables, net of all controls, by equal-size bins. The lines on the graphs show the predicted outcomes (Gaussian kernel, local polynomial smoothing). The confidence intervals are constructed by performing a block bootstrap at the level of the clusters.



Coefficients on each graph come from a single regression; group M has countries with missing GICI index

FIGURE VI

The Effect of 3G Coverage on Government Approval by Overall Corruption Level

The graphs present the coefficients on the interactions between regional 3G coverage and dummies for the 13 groups of countries, grouped by the overall level of corruption (i.e., mean GICI over 2000–2017) with 8 countries in each group. Group M has all 12 countries with missing GICI data. The graphs also present 90% confidence intervals, that are calculated from standard errors, corrected for two-way clusters at the subnational-district level (to account for correlation over time) and at the level of the countries in each year (to account for within-country-year correlation).

TABLE VI
THE RELATIONSHIP BETWEEN ACTUAL AND PERCEIVED CORRUPTION

Dep. var:	Perception of no corruption in government					
	All (1)	Rural (2)	All (3)	Rural (4)	All (5)	Rural (6)
Regional 3G coverage × actual corruption incidents	-0.081*** (0.025)	-0.101*** (0.030)	-0.059** (0.024)	-0.062** (0.029)		
Regional 3G coverage × actual corruption incidents × country with below-median overall corruption					-0.222*** (0.035)	-0.243*** (0.041)
Regional 3G coverage × actual corruption incidents × country with above-median overall corruption					-0.030 (0.020)	-0.039* (0.024)
Regional 3G coverage	-0.014 (0.016)	-0.025 (0.019)	-0.019 (0.016)	-0.037* (0.020)		
Regional 3G coverage × country with below-median overall corruption					0.005 (0.021)	0.002 (0.026)
Regional 3G coverage × country with above-median overall corruption					-0.033* (0.019)	-0.057*** (0.020)
Actual corruption incidents	-0.013 (0.010)	-0.013 (0.012)	-0.017* (0.010)	-0.021* (0.012)	-0.015 (0.010)	-0.017 (0.012)
Observations	691,872	414,346	581,944	354,966	691,872	414,346
Subnational region and year FEs	0.226	0.192	0.151	0.126	0.227	0.193
Baseline controls	✓	✓	✓	✓	✓	✓
Sample excludes observations with zero corruption incidents	✓	✓	✓	✓	✓	✓

Notes: *** $p < .01$, ** $p < .05$, * $p < .1$. The outcome variable is a dummy for the perception that there is no corruption in government. Actual corruption incidents stand for the DII's Global Incidents of Corruption Index (GICI). The unit of observation is an individual. Unreported controls include age, age squared, gender, marital status, dummies for high school and university education, employment status, urban status, the regional average level of income, the log of the country's GDP per capita, the country's unemployment rate, and dummies for democracy status. Standard errors in parentheses are corrected for two-way clusters at the level of the subnational regions (to account for correlation over time) and at the level of the countries in each year (to account for within-country-year correlation).

- Table VII uses the **Panama Papers**. The coefficient on 3G interacted with Panama Papers entities per capita is about **-0.035**; the implied difference-in-differences is about **5 percentage points** for a one-standard-deviation cross-country difference in Panama Papers exposure.
- Figure VII plots the marginal effect of actual corruption on perceived corruption as 3G coverage rises and shows a monotonic steepening.

TABLE VII
3G COVERAGE, THE NUMBER OF ENTITIES IN THE PANAMA PAPERS, AND PERCEIVED CORRUPTION

Dep. var.: Countries in the sample:	Perception of no corruption in government					
	All countries			Excluding low-income countries		
	(1)	(2)	(3)	(4)	(5)	(6)
Regional 3G coverage × × number of Panama Papers entities per capita	-0.035** (0.014)					
× number of Panama Papers entities per capita × before Panama Papers		-0.031** (0.014)	-0.033** (0.014)			
× number of Panama Papers entities per capita × after Panama Papers		-0.037** (0.018)	-0.048*** (0.017)			
× I[Top 10% of countries by Panama Papers entities per capita] × before Panama Papers				-0.045 (0.033)		
× I[Top 10% of countries by Panama Papers entities per capita] × after Panama Papers				-0.100** (0.040)		
× number of Panama Papers entities × before Panama Papers					-0.012*** (0.004)	
× number of Panama Papers entities × after Panama Papers					-0.017*** (0.005)	
× I[Top 10% of countries by Panama Papers entities] × before Panama Papers						-0.092*** (0.028)
× I[Top 10% of countries by Panama Papers entities] × after Panama Papers						-0.174*** (0.038)

TABLE VII
CONTINUED

Dep. var.: Countries in the sample:	Perception of no corruption in government					
	All countries			Excluding low-income countries		
	(1)	(2)	(3)	(4)	(5)	(6)
Regional 3G coverage	-0.027* (0.014)	-0.017 (0.013)	-0.015 (0.014)	-0.015 (0.014)	-0.012 (0.014)	-0.008 (0.014)
Regional 3G coverage × after Panama Papers		-0.011 (0.014)	-0.001 (0.014)	0.003 (0.014)	-0.002 (0.014)	0.000 (0.015)
Observations	722,768	722,768	620,827	620,827	620,827	620,827
R-squared	0.225	0.226	0.232	0.232	0.232	0.232
p-value $\beta(\text{Before Panama Papers}) = \beta(\text{After Panama Papers})$		.490	.055*	.058*	.073*	.0095***
Subnational region and year FEs	✓	✓	✓	✓	✓	✓
Baseline controls	✓	✓	✓	✓	✓	✓
All lower-level interactions	✓	✓	✓	✓	✓	✓
Interactions of 3G and regional income	✓	✓	✓	✓	✓	✓

Notes. *** $p < .01$, ** $p < .05$, * $p < .1$. The outcome variable is a dummy for the perception that there is no corruption in government. "Number of Panama Papers entities" is the number of entities from a country in the Panama Papers. "Number of Panama Papers entities per capita" is the number of entities from a country in the Panama Papers per 1,000 inhabitants. "Before Panama Papers" and "After Panama Papers" are dummies indicating whether the GWP interview took place before or after the release of the Panama Papers to the public. The unit of observation is an individual. Controls include age, age squared, gender, marital status, dummies for high school and university education, employment status, urban status, the region's average level of income, the log of the country's GDP per capita, the country's unemployment rate, dummies for democracy status, the Freedom of the Press score, and the interactions of regional 3G coverage with the region's average level of income. Standard errors in parentheses are corrected for two-way clusters at the level of the subnational regions (to account for correlation over time) and at the level of the countries in each year (to account for within-country-year correlation).

Economic message. The strongest mechanism evidence is that 3G helps citizens learn about actual corruption.

5.7 Table VIII, Table IX, and Figure VIII: electoral implications in Europe

Read:

- Table VIII shows that 3G expansion reduces incumbent electoral support. The coefficient on district 3G coverage is about **-0.089** for the ruling party.
- Using the average district increase in 3G coverage of **0.53**, the implied effect is a **4.7 percentage point** reduction in incumbent vote share over a decade.
- Turnout also falls: the coefficient is about **-0.038**, implying roughly a **2 percentage point** turnout decline.
- Figure VIII shows a clean negative relationship between district 3G expansion and incumbent vote share.
- Table IX shows that among opposition parties, the gains go to **populists**. The coefficient is about **0.086** for right-wing populists and **0.067** for left-wing populists. These imply increases of about **4.6** and **3.6 percentage points**, respectively, for an average decade-long rise in 3G coverage.
- The coefficient for all populists together is about **0.115**, implying roughly a **6.1 percentage point** gain.
- There is essentially **no gain** for Green parties or nonpopulist opposition parties.

Economic message. In Europe, the political consequence of 3G was not a uniform shift to “the opposition.” It was a shift away from incumbents and toward antiestablishment populists.

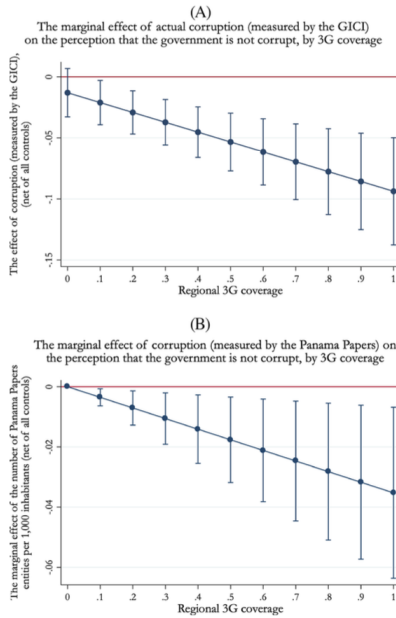


FIGURE VII

3G Coverage and Actual and Perceived Corruption

The outcome variable is a dummy for the perception that there is no corruption in government. In Panel A, the explanatory variables are: regional 3G coverage, actual corruption incidents (GICI), their interaction term, as well as all the baseline controls, including region and year fixed effects (Table VI, column (1)). In Panel B, the explanatory variables are: regional 3G coverage, the interaction term of regional 3G coverage and the number of entities in the Panama Papers per 1,000 people, the interaction of regional 3G coverage with regional income, as well as all the baseline controls, including region and year fixed effects (Table VII, column (1)). The graphs present the marginal effects of an increase in actual corruption (measured by the GICI and the Panama Papers) on the perception of corruption. The graphs also present 95% confidence intervals, calculated from standard errors, corrected for two-way clusters at the level of the subnational districts (to account for correlation over time) and at the level of the countries in each year (to account for within-country-year correlation). The difference in the shape of the confidence intervals in the two graphs comes from the fact that the GICI varies both across countries and over time, whereas the Panama Papers provide information on countries at one point in time.

TABLE VII
THE EFFECT OF 3G COVERAGE ON INCUMBENT'S ELECTORAL PERFORMANCE IN ECUADOR

Dep. var.	Vote share of:				
	The 2 parties from the last election	Ruling party (the party of the previous incumbent)	Opposition parties (they are among the 2 parties from the last election)	Turnout	
Unit of observation:	District-year	District-year	District-year	District-year	District-year
District 3G coverage	-0.008** (0.003)	-0.008** (0.003)	-0.008** (0.003)	-0.008** (0.003)	-0.008** (0.003)
District 3G coverage × population party			-0.008** (0.003)		
Observations	1,204	1,204	1,204	1,204	1,204
Adjusted R-squared	0.008	0.007	0.008	0.008	0.008
Mean dep. var.	0.501	0.501	0.501	0.501	0.501
Mean 3G coverage	0.049	0.049	0.049	0.049	0.049
District and year FE	✓	✓	✓	✓	✓
Incumbent in district and year FE	✓	✓	✓	✓	✓
Random controls	✓	✓	✓	✓	✓
Full equation without population coverage by 2 in the last election					

TABLE IX
THE EFFECT OF 3G COVERAGE ON THE GOVERNMENT'S ELECTORAL PERFORMANCE IN ECUADOR

Dep. var.	Vote share of:					
	Right-wing populists	Left-wing populists	Other populists	All populists	Other parties	Nonpopulist opposition parties
Unit of observation:	District-year	District-year	District-year	District-year	District-year	District-year
District 3G coverage	0.008** (0.003)	0.007** (0.003)	-0.008 (0.004)	0.007** (0.003)	-0.007 (0.004)	-0.008 (0.004)
Observations	1,204	1,204	1,204	1,204	1,204	1,204
Adjusted R-squared	0.008	0.007	0.004	0.007	0.007	0.004
Mean dep. var.	0.501	0.501	0.501	0.501	0.501	0.501
Mean 3G coverage	0.049	0.049	0.049	0.049	0.049	0.049
District and year FE	✓	✓	✓	✓	✓	✓
Right-wing populists by district and year FE	✓	✓	✓	✓	✓	✓
Left-wing populists by district and year FE	✓	✓	✓	✓	✓	✓
Other populists by district and year FE	✓	✓	✓	✓	✓	✓
Random controls	✓	✓	✓	✓	✓	✓
Full equation with populists in power						

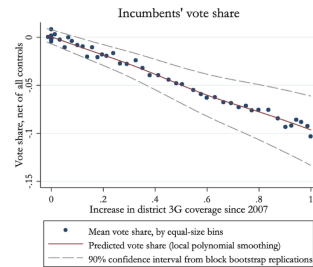


FIGURE VIII

Electoral Implications of the Expansion of 3G Coverage for Incumbents

The figure illustrates the results presented in Table VIII, column (2). The dots represent the vote shares, net of all controls, by equal-size bins. The solid line on the graphs shows the predicted vote shares (Gaussian kernel, local polynomial smoothing). The 90% confidence intervals are constructed by performing a block bootstrap at the level of the clusters.

6 Short summary

- The paper studies the political effects of 3G mobile internet using global subnational data.
- The main finding is that more 3G lowers confidence in government and increases perceived corruption.
- The effect is strongest where the internet is not censored and where traditional media are more censored.
- The paper provides unusually direct mechanism evidence: 3G strengthens the link between actual corruption and perceived corruption.
- The results are robust to country-year fixed effects, pretrend checks, event studies, a 2G placebo, IV based on lightning strikes, and placebo outcomes.
- In Europe, 3G reduces incumbent vote shares and raises populist vote shares.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that the expansion of mobile broadband internet reduced government approval around the world and had tangible electoral consequences in Europe. The effect is not merely correlational in levels; it survives demanding within-region panel designs and several distinct credibility checks.

7.2 Economic intuition

3G matters because it changes who can access political information, how quickly they can access it, and what formats that information takes. When the internet is uncensored, citizens become more aware of corruption and less trusting of incumbent governments. Where traditional media are controlled, the effect is even stronger because 3G becomes the main route to independent information.

7.3 Takeaway

The paper's main message is that mobile broadband is politically transformative. It can improve accountability by exposing corruption, but it can also destabilize incumbents and channel voter frustration toward populist opposition. In that sense, the diffusion of 3G is both an information shock and a political shock.